Review of Mathematical Theory

1-7

Theory of Computation

Let us discuss each one by one -

1) Negation

This proposition is also called NOT proposition.It is denoted by I or ~ or having horizontal bar on letter. If S is a proposition then S defines NOTS.The truth table for this connective is

S	15
	F
T	
	T
F	

ii) Conjunction (AND

If A and B are two propositions then conjunction of A and B is denoted by A^B. The

o piop		
nective is		A^B
	B	
A		T
	T	
T		F
	F	
T		F
	T	
F		F
	F	
F		

truth table for this connective is

Lf A and B are two propositions then disjunction of A and B is denoted by AvB.The

ii) Disjunction (OR)

vo propos		
nnective is		AvB
	B	
A		T
	T.	
T		T
	F	
T		T
	T	
F		F

vo propos
F
F
nositi

truth table for this connective is

f A and B are two propositions then if A then B is a proposition.It can be denoted

Iv) Implication

byABorA = B

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The truth table for this connective is Properti

A	B	AB
T	T	T
T	F	F
F	T	T
F	F	T

For example: Find truth value for following proposition.

If an apple is not green then it must be red.

Let A denotes An apple is not green and B denotes It must be redThe statement

A is false and B is true. Hence the proposition is true

v)If and only if

If A and B are two propositions then A if and only if B.It can be denoted as AB

The truth table will be Equiv

A	B	AB
T	T	T
T	F	F
F	T	F
F	F	T

For example: I will go to the school if and only if it is not raining.Here statement A is I will go to the school. The statement B is It is not raining. Thus the resultant proposition is false.

1.5 Relations

Relasonship is a major aspect between two objects even this is true in our real life.

One oyject can be related with the other object by a some relation.Then those two objects forma pair based on this certain relationship. Clos

Definition The relation R is a collection for the set S which represents the pair of elements.

For example (a b is in R. We can represent their relation as a R b. The first component ol each pair is chosen from a set called domain and second component of each pair is chosen from a set called range

Properties of Relations

A relation R on set S is

1. Reflexive if iRi for all i in S.
2. Irreflexive if iRi is false for all i in S.
3. Transitive if iRj and jRk imply iRk

Symmetric if iRj implies jRi.

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5. Asymmetric if iRj implies that jRi is false.

Every asymmetric relation must be irreflexive

For example: if A = ab| then

reflexive relation R can be = aab, b
irreflexive relation R can be=|a, b1
transitive relation R can be = ab).baa|
symmetric relation R can be=|a, b).ba
asymmetric relation R can be =|a, b

Equivalent Relation

A relation is said to be equivalence relation if it is reflexive, symmetric and transitive, over some set S.

Suppose R is a set of relations and S is the set of elements.

For example: S is the set of lines in a plane and R is the relation of lines intersecting to each other.

A relation is said to be antisymmetric if a b)e R and b, a e R then a =a.

Example 1.5: Determine whether R is equivalence relation or not where

A = 0121R = 00)1.0)1, 12, 221

Solution The R is reflexive because 0, 0).1, 1)2.2E R.

Where as R is not symmetric because (0, 1) is not in R whereas (1, 0) is in R.

Hence the R is not a equivalence relation.

Closures of Relations

Sometimes when the relation R is given it may not be reflexive or transitive

By adding some pairs we make the relation as reflexive or transitive.

For exampleLetab)bc)a, abb